NetworkLab Arena: A Multiplayer Aim Trainer for Studying Network Latency and Netcode Synchronization

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1. Motivation and Personal Context

Modern competitive online games demand real-time state synchronization across geographically distributed clients. As an active player of competitive first-person shooters like *Valorant*, I frequently experience how minor network fluctuations degrade gameplay. A sudden ping spike from 20 ms to 70 ms or temporary packet loss often results in a shot that clearly landed on the client screen failing to register on the authoritative game server. This discrepancy between client-rendered state and server-authoritative state creates significant frustration and compromises competitive fairness.

While commercial game engines employ latency compensation algorithms such as server-side backward reconciliation and client prediction, evaluating these mechanisms under controlled network conditions remains challenging. Commercial netcode is proprietary, making it difficult to isolate specific variables like tick intervals or interpolation delays. *NetworkLab Arena* addresses this gap by providing an open, lightweight experimental testbed designed to measure spatial and temporal synchronization metrics under controlled network impairments.

2. System Architecture and Implementation Plan

NetworkLab Arena consists of a custom two-dimensional socket server built in Node.js operating at a 60 Hz tick rate, paired with an interactive browser-based HTML5 canvas client. The server maintains the authoritative state of all game objects, processes input packets, and broadcasts compressed state updates.

The testbed operates in two configurable netcode modes:

1. *Uncompensated Server Mode*: The server processes incoming shot events against the target's position at the exact moment the packet arrives, exposing the full impact of network delay.
2. *Lag-Compensated Rewind Mode*: The server maintains a short historical position buffer. Upon receiving a shot event timestamped by the client, the server rewinds target locations to match what the player saw at click time.

To evaluate these modes under realistic internet conditions, network impairments will be artificially injected using Linux traffic control (*tc*) and *netem* utilities. This setup enables precise testing across fixed delays (10 ms to 150 ms), random packet loss (1% to 10%), and jitter profiles.

3. Quantitative Evaluation Metrics

To assess netcode quality, *NetworkLab Arena* logs telemetry across three core mathematical metrics:

1. *Spatial Pixel Error Margin (Δpx)*: Measures the Euclidean distance between the client-rendered target position (x_c, y_c) at click time and the authoritative server position (x_s, y_s) when the event is processed:

$$\Delta px = \sqrt{(x_c - x_s)^2 + (y_c - y_s)^2}$$

2. *Server Tick Interval (Δt)*: Tracks the elapsed execution time between consecutive server loop iterations to evaluate frame stability under load:
$$\Delta t = t_n - t_{n-1}$$

3. *Tick Jitter ($\sigma_{\Delta t}$)*: Calculates the standard deviation of tick intervals across N sample ticks to quantify frame timing consistency:
$$\sigma_{\Delta t} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\Delta t_i - \bar{\Delta} t)^2}$$

4. Expected Outcomes

The primary outcome of *NetworkLab Arena* is an empirical evaluation showing how spatial drift increases with network latency, and quantifying the accuracy recovery provided by lag compensation. By systematically logging telemetry across various latency and jitter profiles, the project will produce quantitative benchmarks comparing uncompensated and lag-compensated netcode modes under controlled network impairments.

5. References

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